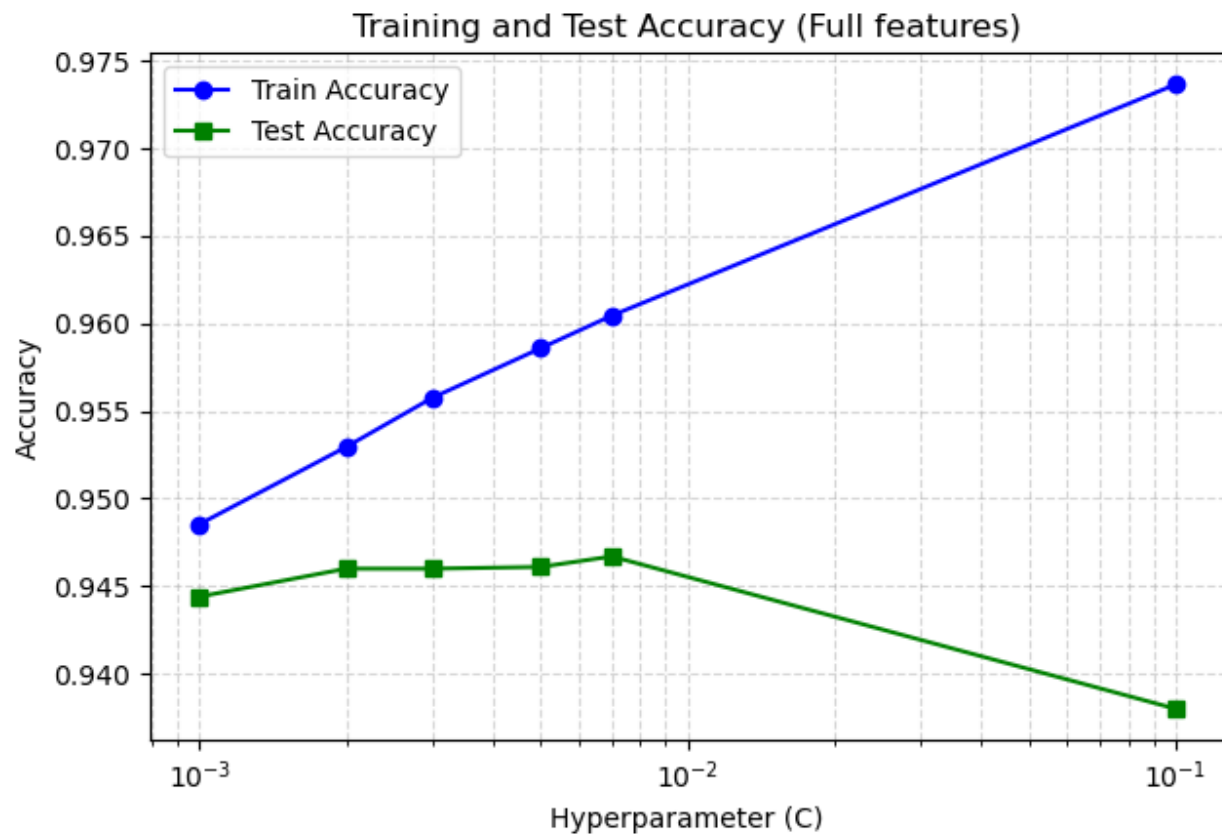


Experiment and results

MNIST dataset

Linear SVC (Tuning C)

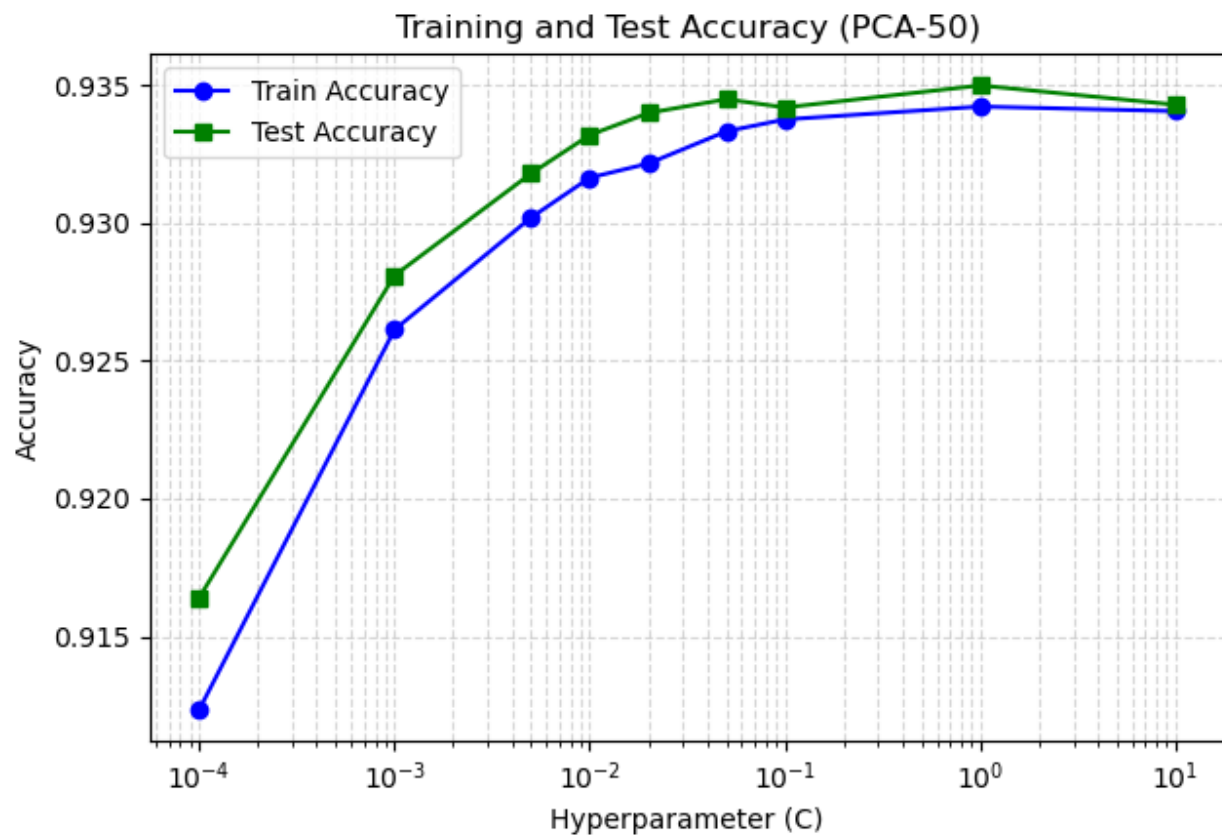
Without dimensionality reduction



C	Training Time	Training Accuracy	Test Accuracy
0.001	157.73118	0.948533	0.9444
0.002	132.326683	0.95295	0.946
0.003	131.465942	0.95575	0.946
0.005	134.293067	0.958583	0.9461
0.007	134.292979	0.96045	0.9467
0.1	161.697288	0.97365	0.938

C = 0.02 is optimal

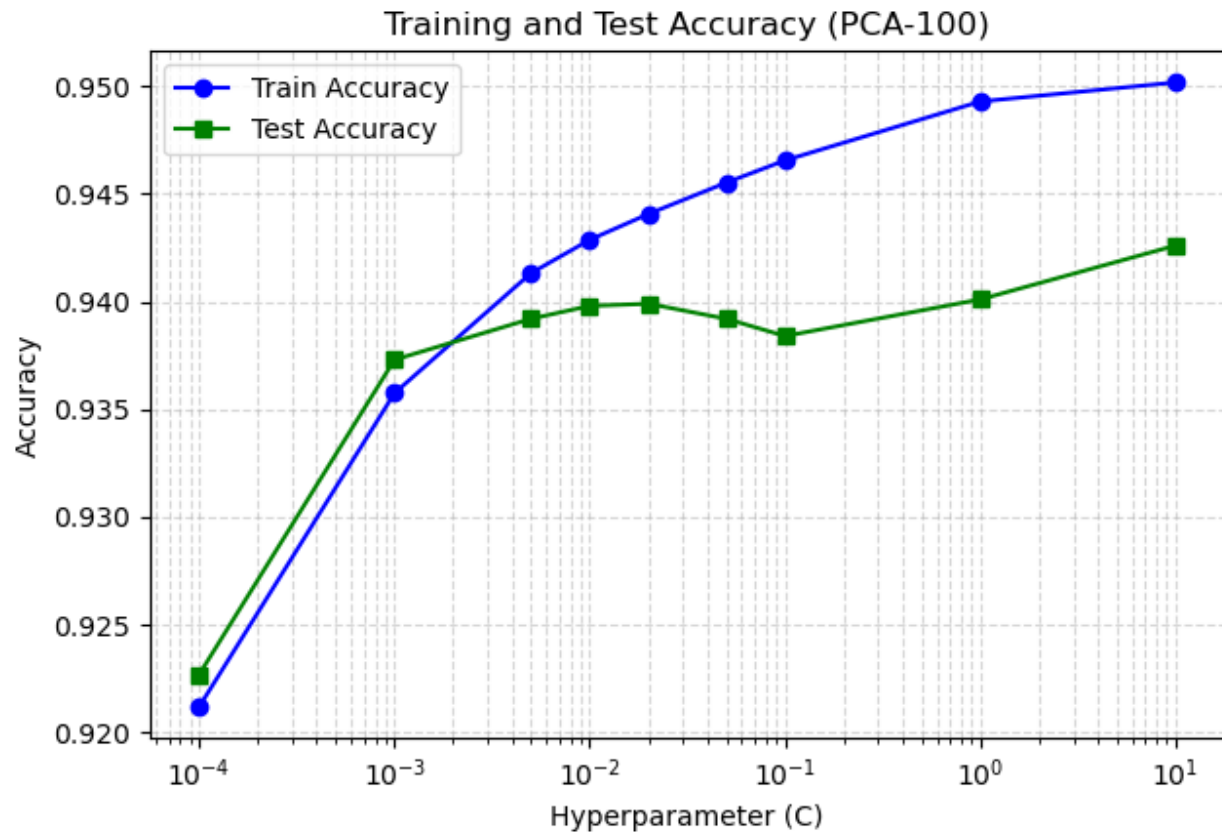
PCA (Dim = 50)



C = 1 looks to be the best.

	Training Time	Training Accuracy	Test Accuracy
0.0001	29.768446	0.912333	0.9164
0.001	14.415382	0.926133	0.9281
0.005	14.051898	0.9302	0.9318
0.01	14.120615	0.93165	0.9332
0.02	14.237202	0.932167	0.934
0.05	17.481171	0.933333	0.9345
0.1	19.428435	0.933767	0.9342
1	81.220042	0.934233	0.935
10	588.948056	0.934067	0.9343

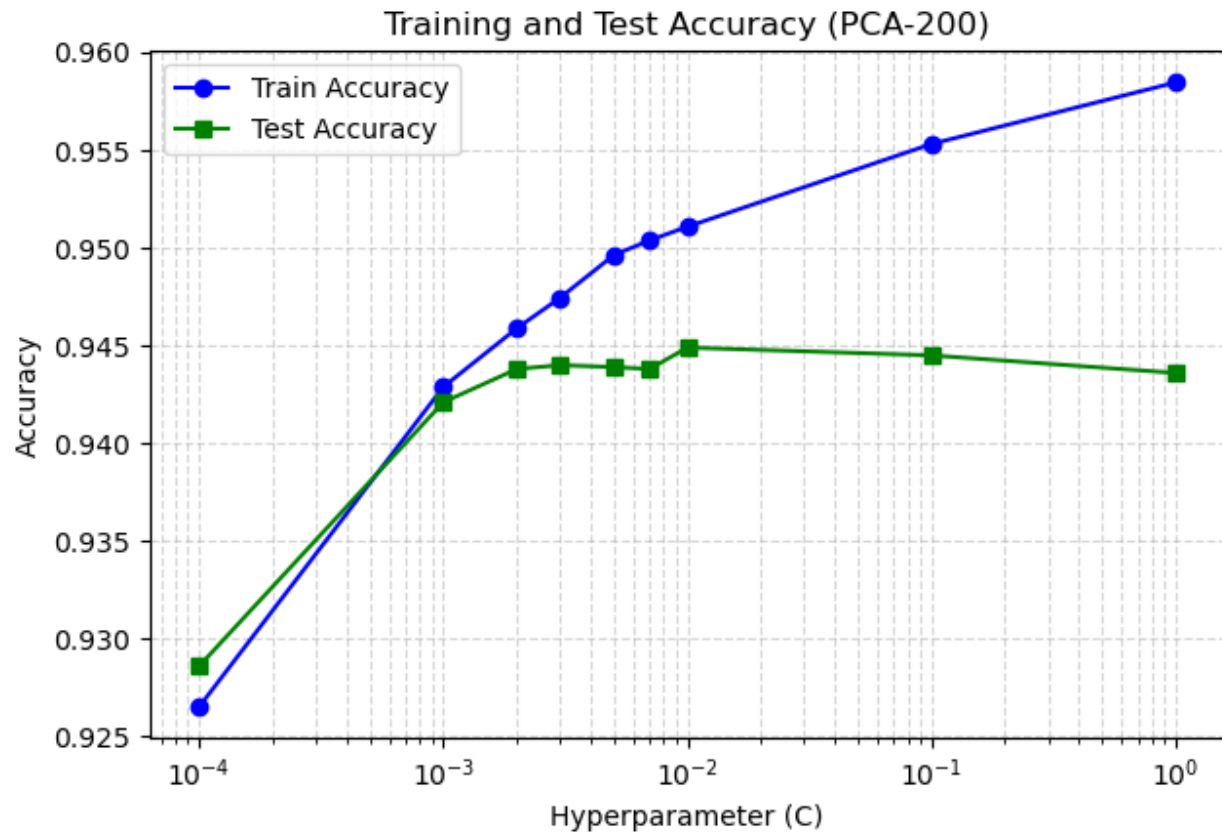
PCA (Dim = 100)



	Training Time	Training Accuracy	Test Accuracy
0.0001	43.068827	0.9212	0.9227
0.001	19.935494	0.93575	0.9373
0.005	15.271159	0.941317	0.9392
0.01	14.379993	0.942867	0.9398
0.02	15.674328	0.944067	0.9399
0.05	18.078031	0.945517	0.9392
0.1	23.421693	0.94655	0.9384
1	96.342034	0.949283	0.9401
10	824.605545	0.950167	0.9426

C = 0.01 is the best

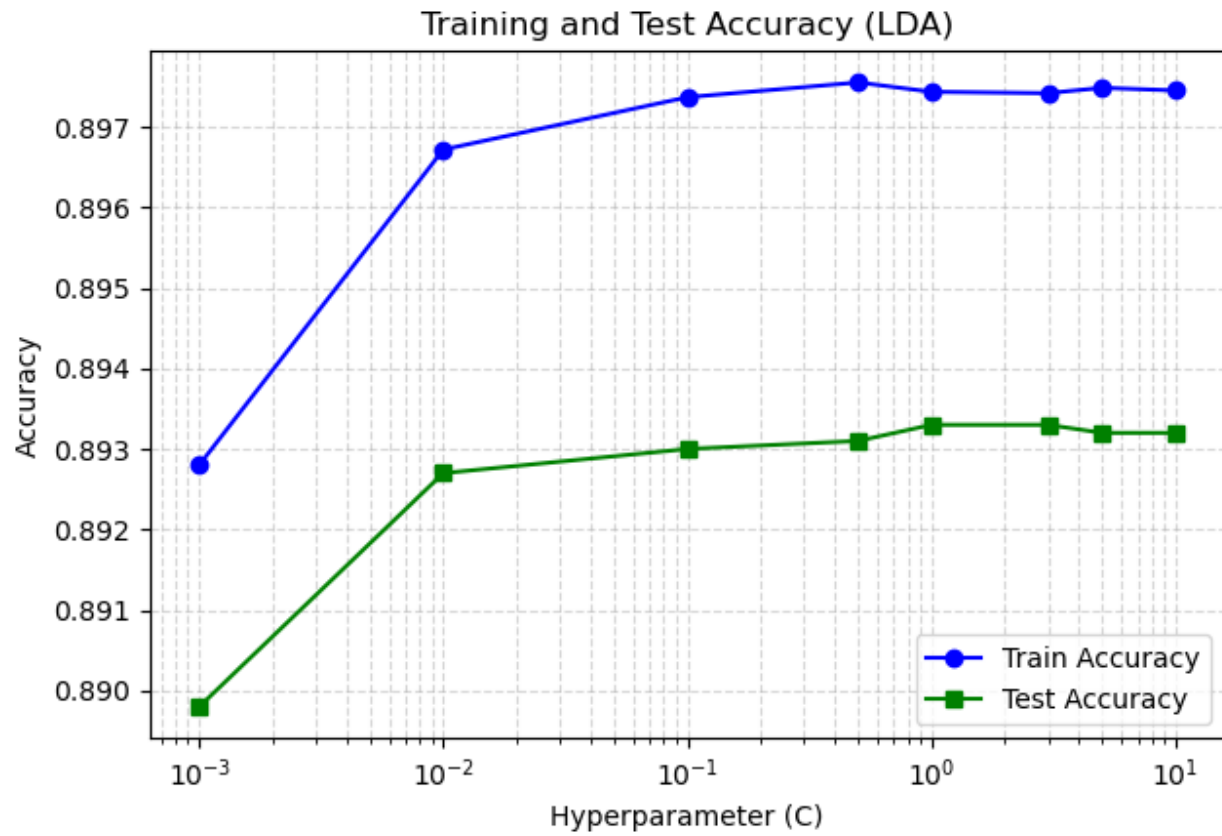
PCA (Dim = 200)



C=0.005 seems to be the best

	Training Time	Training Accuracy	Test Accuracy
0.0001	105.831126	0.926517	0.9286
0.001	48.096166	0.942883	0.9421
0.002	38.54958	0.945883	0.9438
0.003	35.798097	0.947433	0.944
0.005	35.078782	0.949633	0.9439
0.007	31.530684	0.950383	0.9438
0.01	33.643679	0.951083	0.9449
0.1	51.411778	0.955317	0.9445
1	161.99293	0.958467	0.9436

LDA

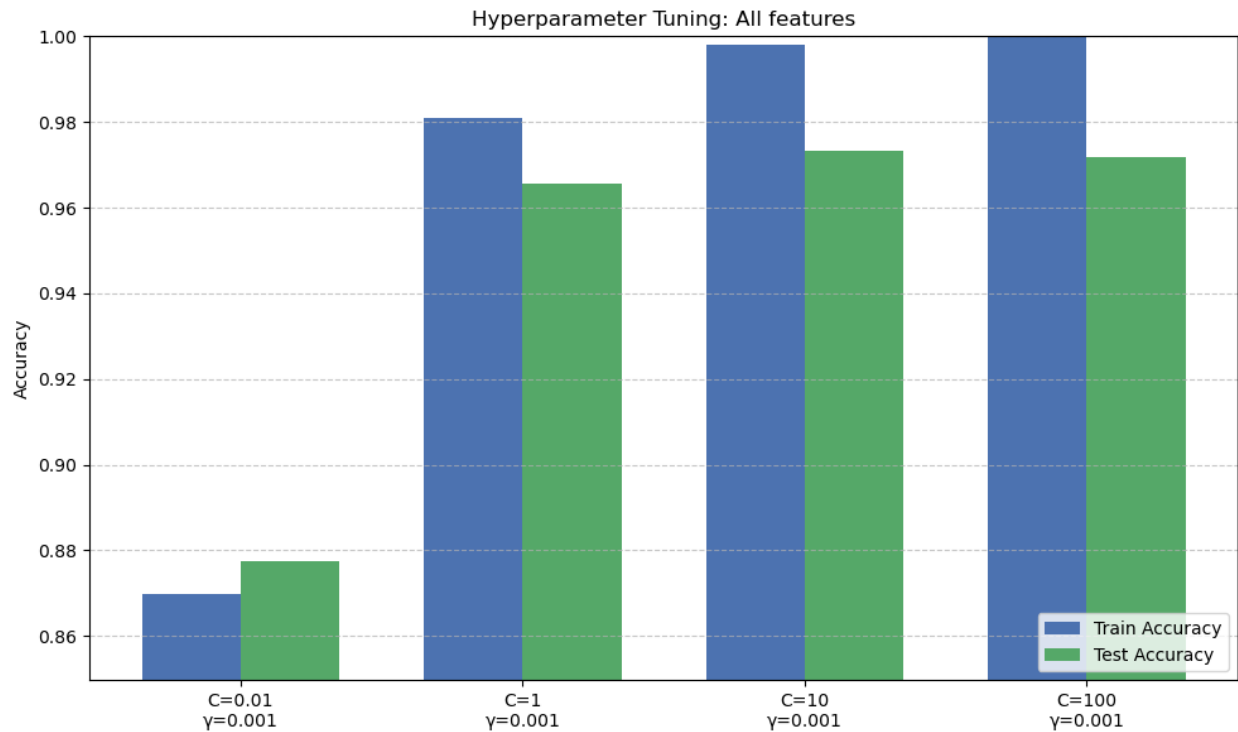


	Training Time	Training Accuracy	Test Accuracy
0.001	20.397919	0.8928	0.8898
0.01	15.872321	0.896717	0.8927
0.1	15.650555	0.897367	0.893
0.5	18.37171	0.89755	0.8931
1	19.348776	0.897433	0.8933
3	26.523438	0.897417	0.8933
5	33.413828	0.897483	0.8932
10	48.142169	0.89745	0.8932

C = 1 is optimal

RBF SVC Tuning

Without Dimensionality Reduction



C	Gamma	Training Time (s)	Training Accuracy	Test Accuracy
0.01	0.001	1235.032199	0.86975	0.8774
1	0.001	233.945355	0.981083	0.9657
10	0.001	192.541993	0.99805	0.9733
100	0.001	202.584364	1	0.9717

PCA-50

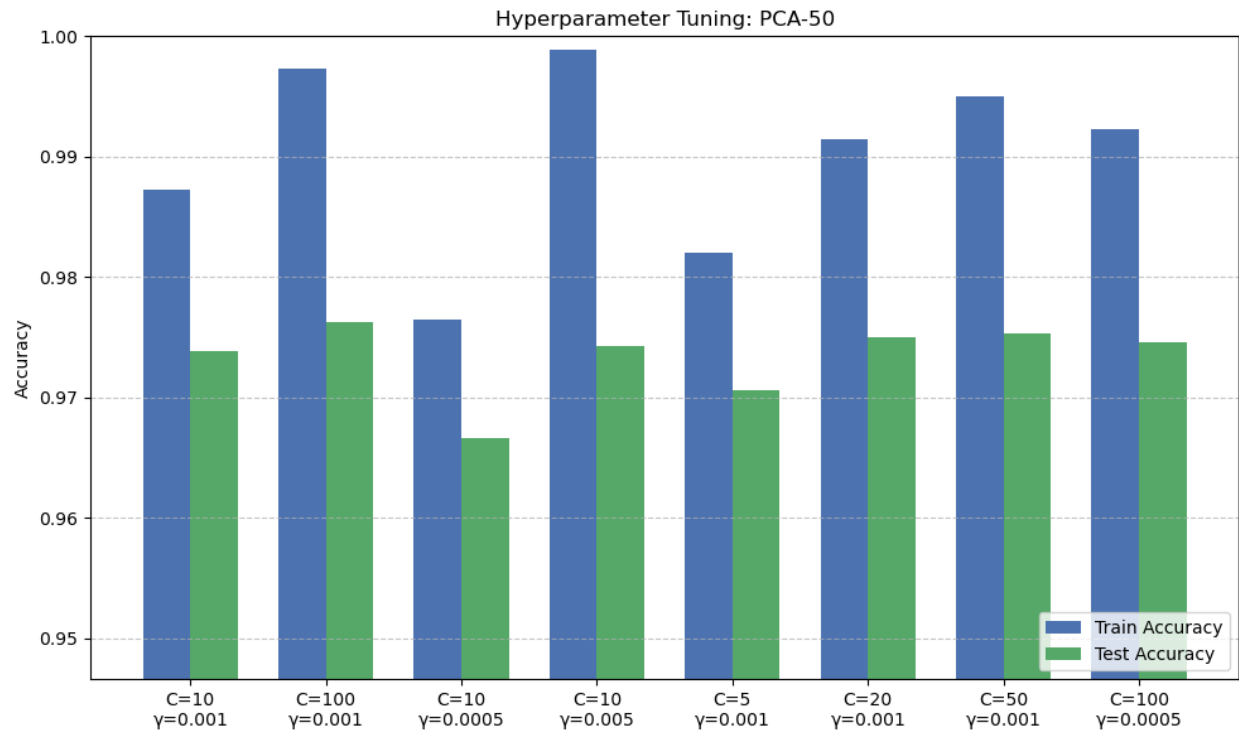
Tried a bunch of large ranged values for C and gamma like:

C = [0.01, 1, 10, 100]

gamma = [0.0001, 0.001, 0.01, 0.1, 1, 10, 100]

Once we established a general idea of good parameters, we ran more fine grained parameter adjustment to find the optimal values.

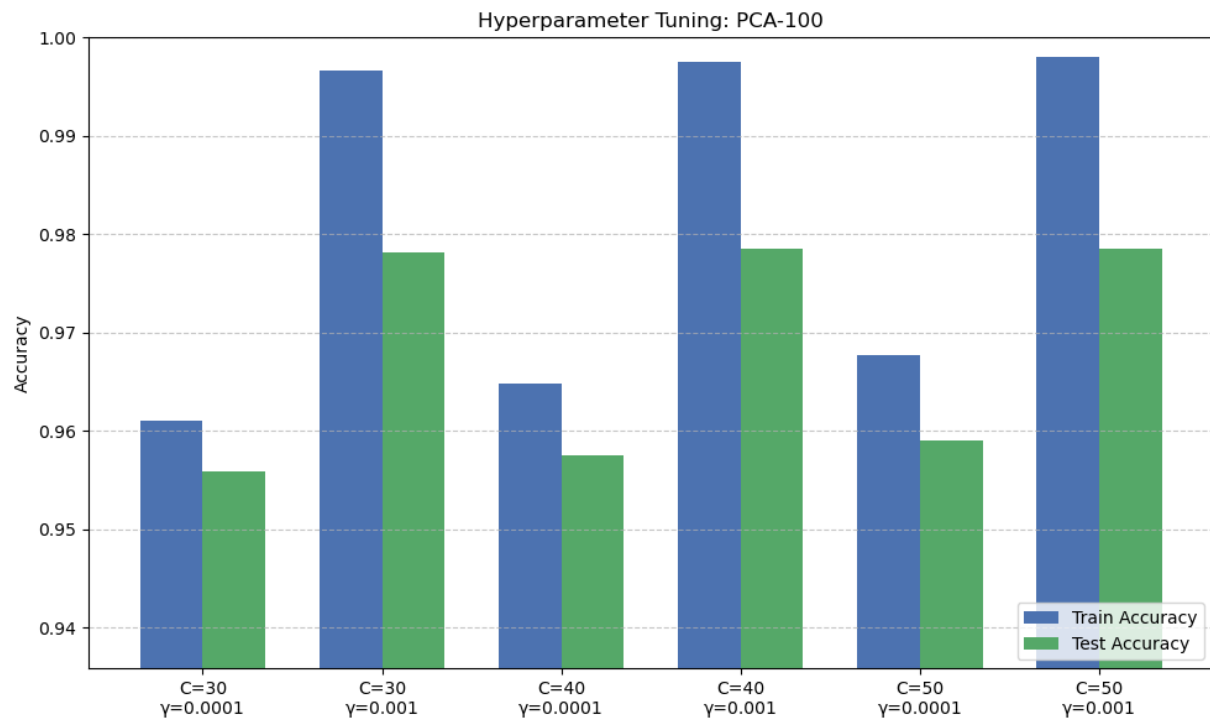
Fine-tuning



C	gamma	Training Time	Training accuracy	Test Accuracy
10	0.001	11.950259	0.987233	0.9738
100	0.001	11.437899	0.9973	0.9763
10	0.0005	12.567419	0.9765	0.9666
10	0.005	25.860326	0.9989	0.9743
5	0.001	12.916175	0.982033	0.9706
20	0.001	13.247008	0.991433	0.975
50	0.001	12.415907	0.99505	0.9753
100	0.0005	11.158599	0.992267	0.9746

C = 100, gamma = 0.001 has the best test accuracy, but the training accuracy is also very high.

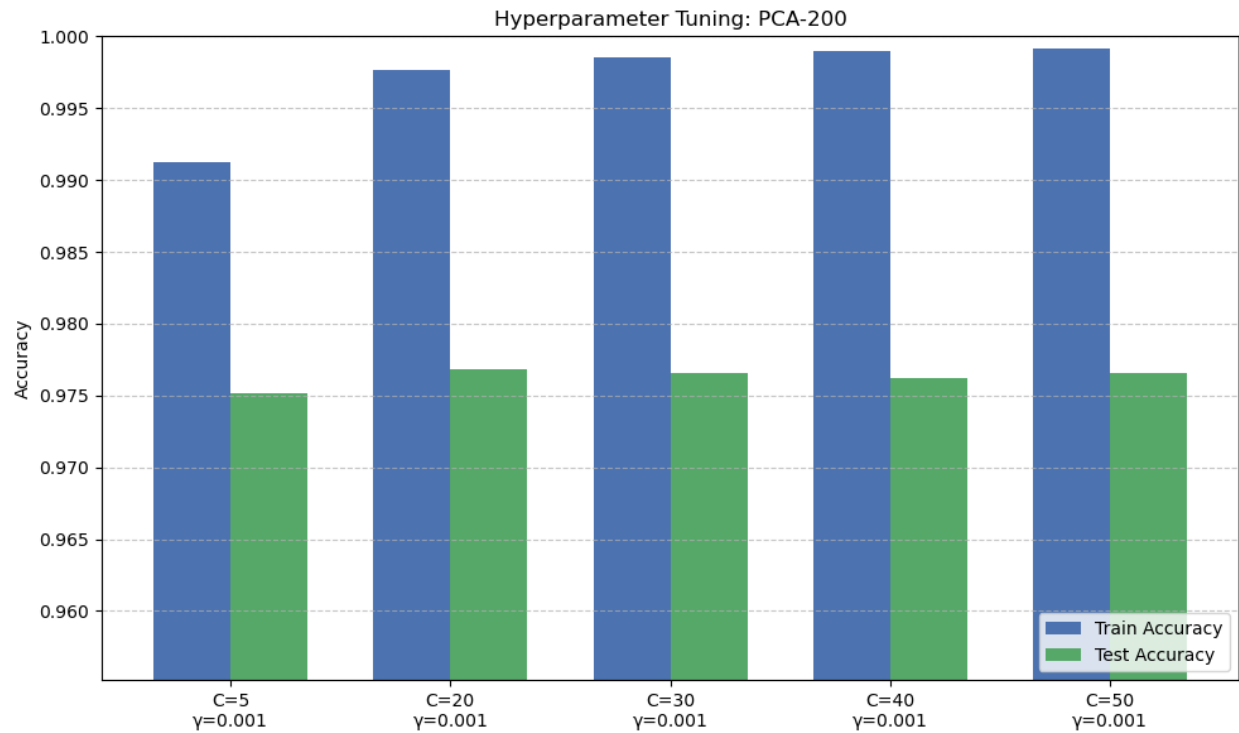
PCA-100



C=40, gamma = 0.001 is the most optimal parameters

C	Gamma	Training Time	Training Accuracy	Test Accuracy
30	0.0001	22.601361	0.96105	0.9559
30	0.001	18.505395	0.99665	0.9782
40	0.0001	26.393991	0.964817	0.9576
40	0.001	23.377848	0.997533	0.9785
50	0.0001	22.192176	0.967767	0.959
50	0.001	17.735109	0.998	0.9785

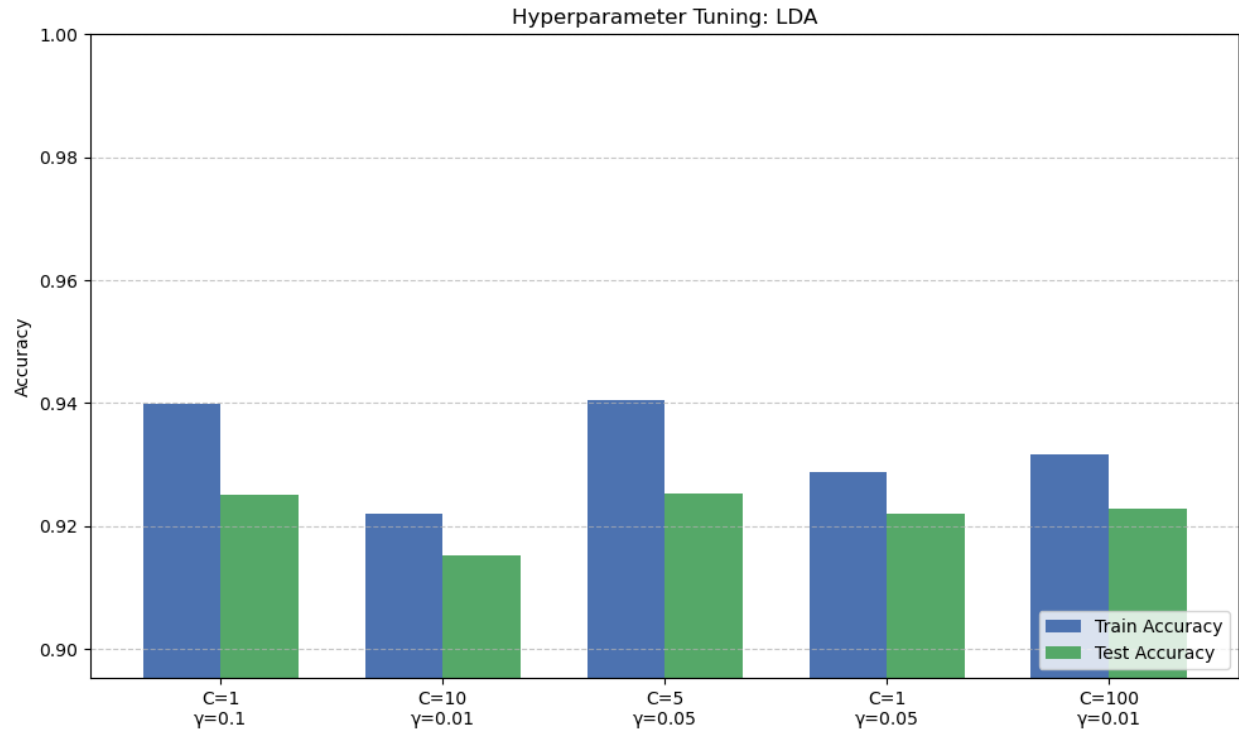
PCA-200



Clearly, C = 20, gamma = 0.001 are the most optimal parameters.

C	Gamma	Training Time	Training Accuracy	Test Accuracy
5	0.001	57.685208	0.991217	0.9752
20	0.001	53.276627	0.997683	0.9768
30	0.001	55.882794	0.998517	0.9766
40	0.001	55.185333	0.998967	0.9762
50	0.001	51.825474	0.99915	0.9766

LDA



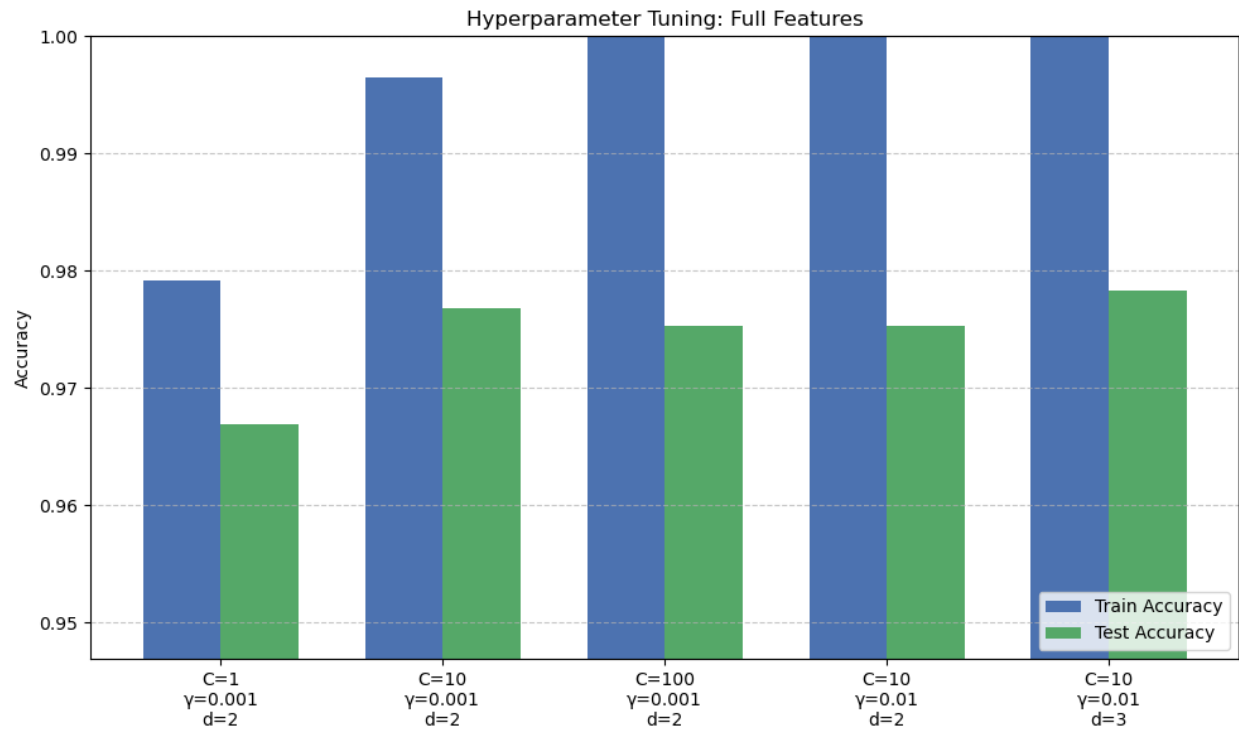
C	Gamma	Training Time (s)	Training Accuracy	Test Accuracy
1	0.1	20.586518	0.9398	0.9252
10	0.01	17.478508	0.921967	0.9153
5	0.05	16.708928	0.940583	0.9254
1	0.05	14.868029	0.928817	0.922
100	0.01	17.853528	0.931767	0.9229

Polynomial SVC

It seems degree 4 has very high overfitting. It had very low test accuracy while the training accuracy was very high. So, we only test degrees 2 and 3 for this dataset.

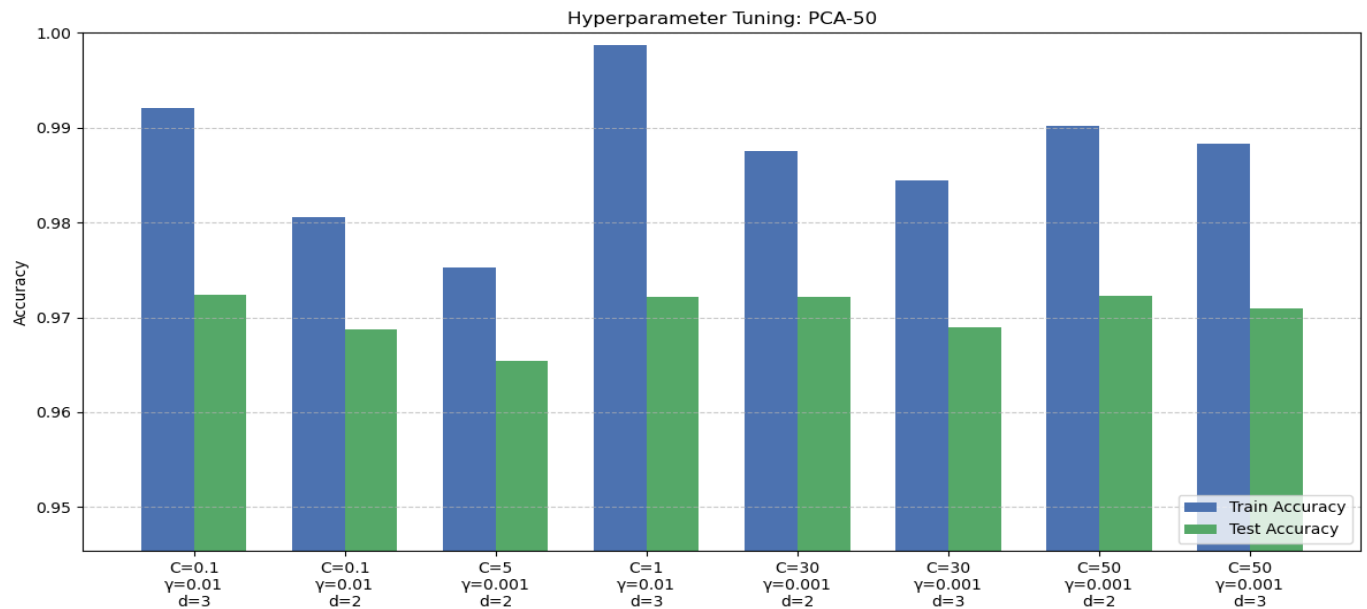
No dimensionality reduction

Initial test showed all degree 3 overfitted the dataset.



C	Gamma	Degree	Training Time	Training Accuracy	Test Accuracy
1	0.001	2	358.473889	0.979167	0.9669
10	0.001	2	195.243409	0.996467	0.9768
100	0.001	2	182.277205	0.99995	0.9753
10	0.01	2	189.829204	1	0.9753
10	0.01	3	298.727221	1	0.9783

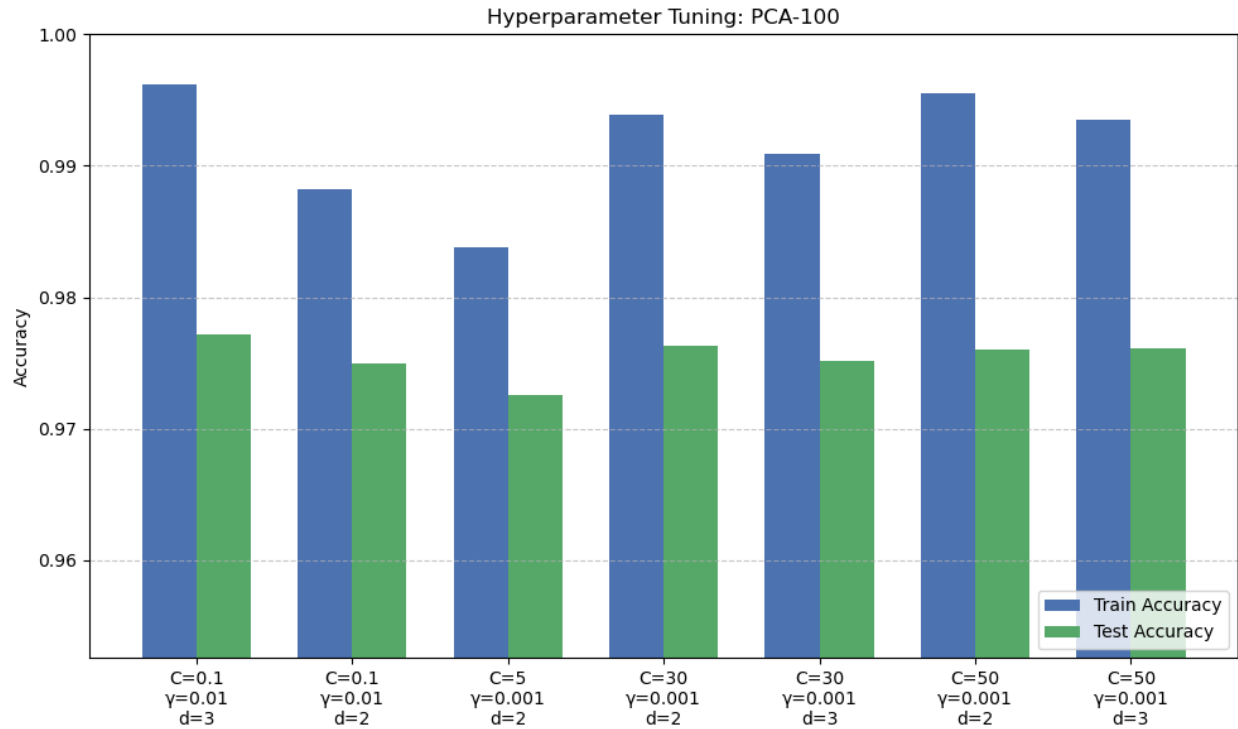
PCA-50



C	Gamma	Degree	Training Time	Training Accuracy	Test Accuracy
0.1	0.01	3	15.954031	0.992017	0.9724
0.1	0.01	2	17.941901	0.980583	0.9687
5	0.001	2	19.837571	0.975267	0.9654
1	0.01	3	13.20148	0.998683	0.9722
30	0.001	2	12.998366	0.987567	0.9722
30	0.001	3	21.027206	0.984433	0.9689
50	0.001	2	11.776544	0.990217	0.9723
50	0.001	3	16.748056	0.988283	0.9709

C=30, gamma = 0.001, degree = 2 would be the optimal value since test accuracy is like the other 2 and training accuracy isn't much higher, which indicates more generalization.

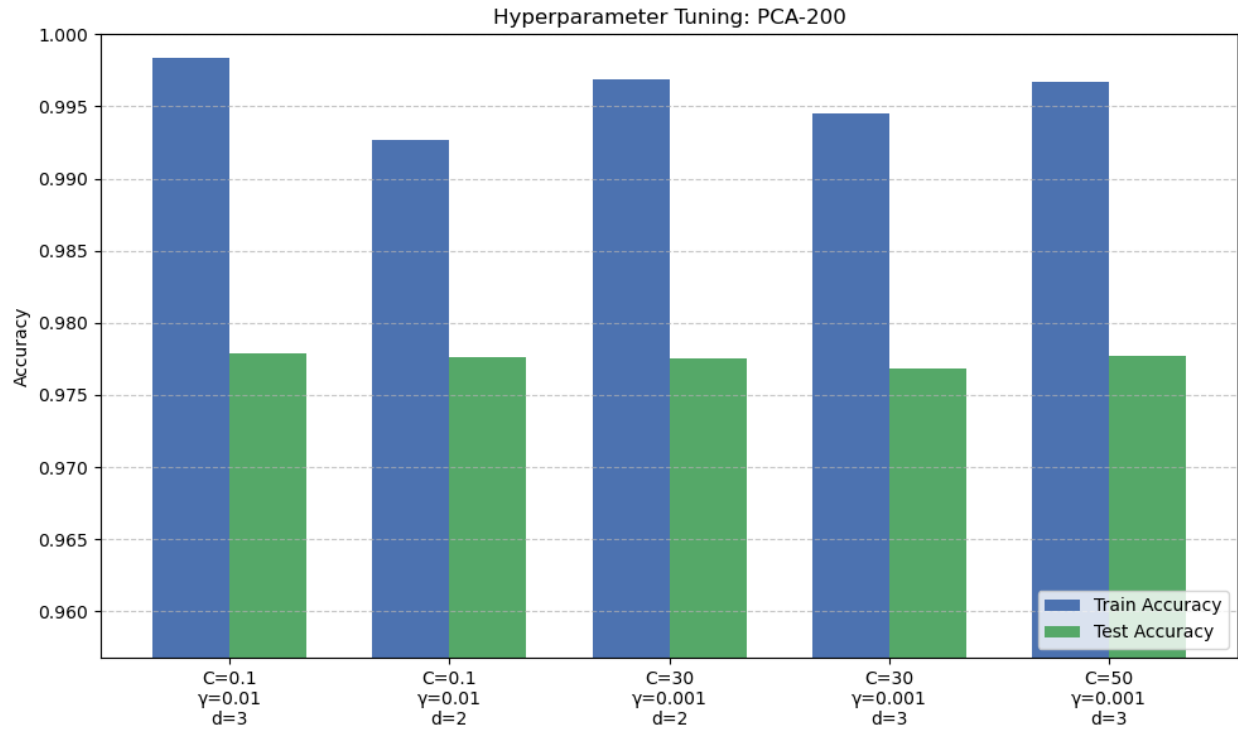
PCA-100



C	Gamma	Degree	Training Time	Training Accuracy	Test Accuracy
0.1	0.01	3	27.55803	0.99615	0.9772
0.1	0.01	2	24.736106	0.988267	0.975
5	0.001	2	32.160097	0.983783	0.9726
30	0.001	2	18.522764	0.993867	0.9763
30	0.001	3	30.558123	0.990917	0.9752
50	0.001	2	16.656505	0.9955	0.976
50	0.001	3	27.409001	0.993533	0.9761

C = 30, gamma = 0.001, Degree = 2 would be the optimal hyperparameters.

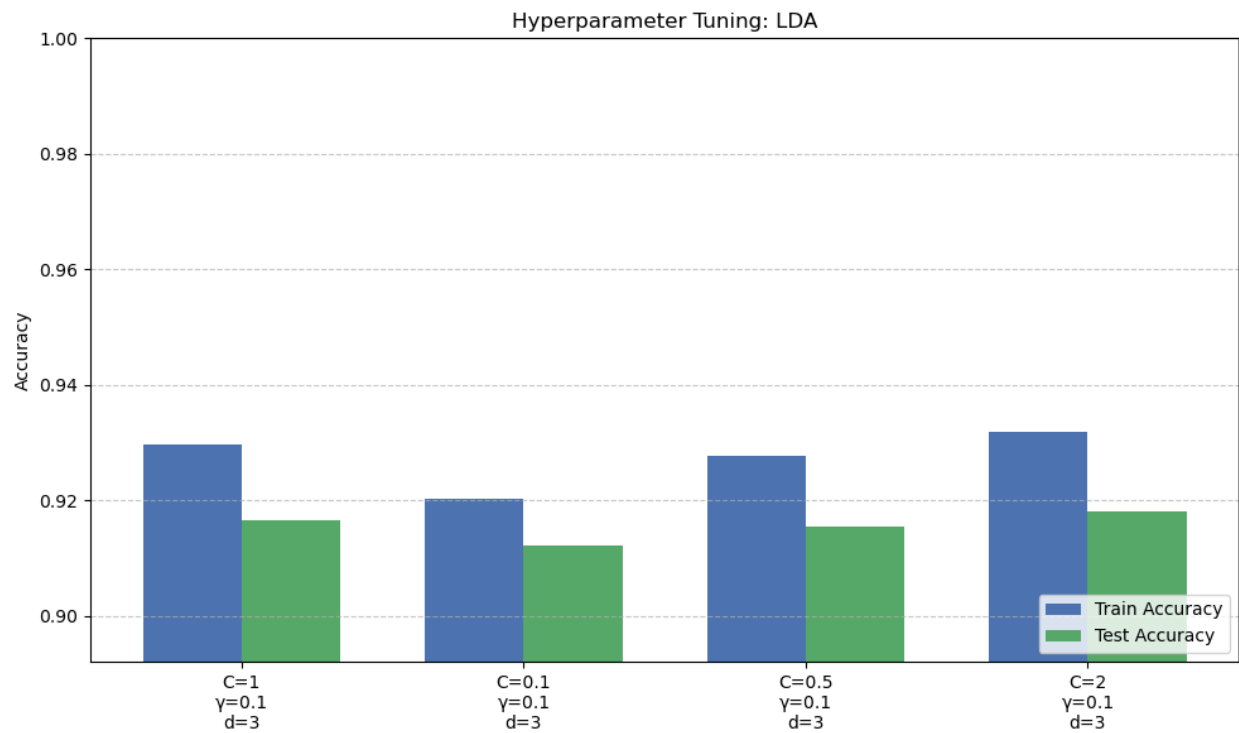
PCA-200



C	Gamma	Degree	Training Time	Training Accuracy	Test Accuracy
0.1	0.01	3	61.867063	0.998367	0.9779
0.1	0.01	2	49.56462	0.99265	0.9776
30	0.001	2	39.011614	0.996917	0.9775
30	0.001	3	66.715055	0.994483	0.9768
50	0.001	3	62.889079	0.996683	0.9777

C = 0.1, gamma = 0.01, degree = 2 seems to be the optimal hyperparameter

LDA



C	Gamma	Degree	Training Time	Training Accuracy	Test Accuracy
1	0.1	3	21.777578	0.929683	0.9166
0.1	0.1	3	21.126786	0.920333	0.9121
0.5	0.1	3	16.845272	0.92765	0.9155
2	0.1	3	19.807027	0.931917	0.918

MNIST ensembling

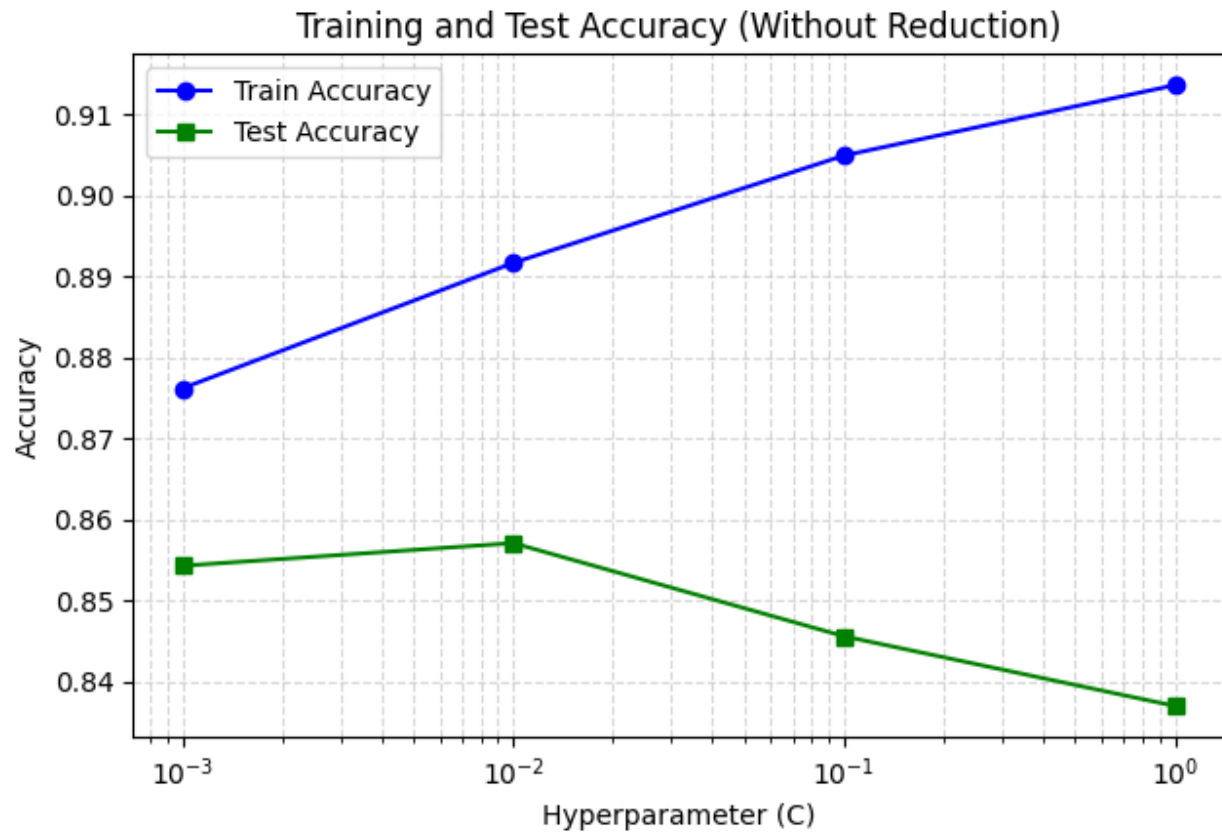
Training time: 15.60067209999761 s

Accuracy: 0.9583, Error: 0.04169999999999996

Fashion MNIST dataset

Linear SVC (Tuning C)

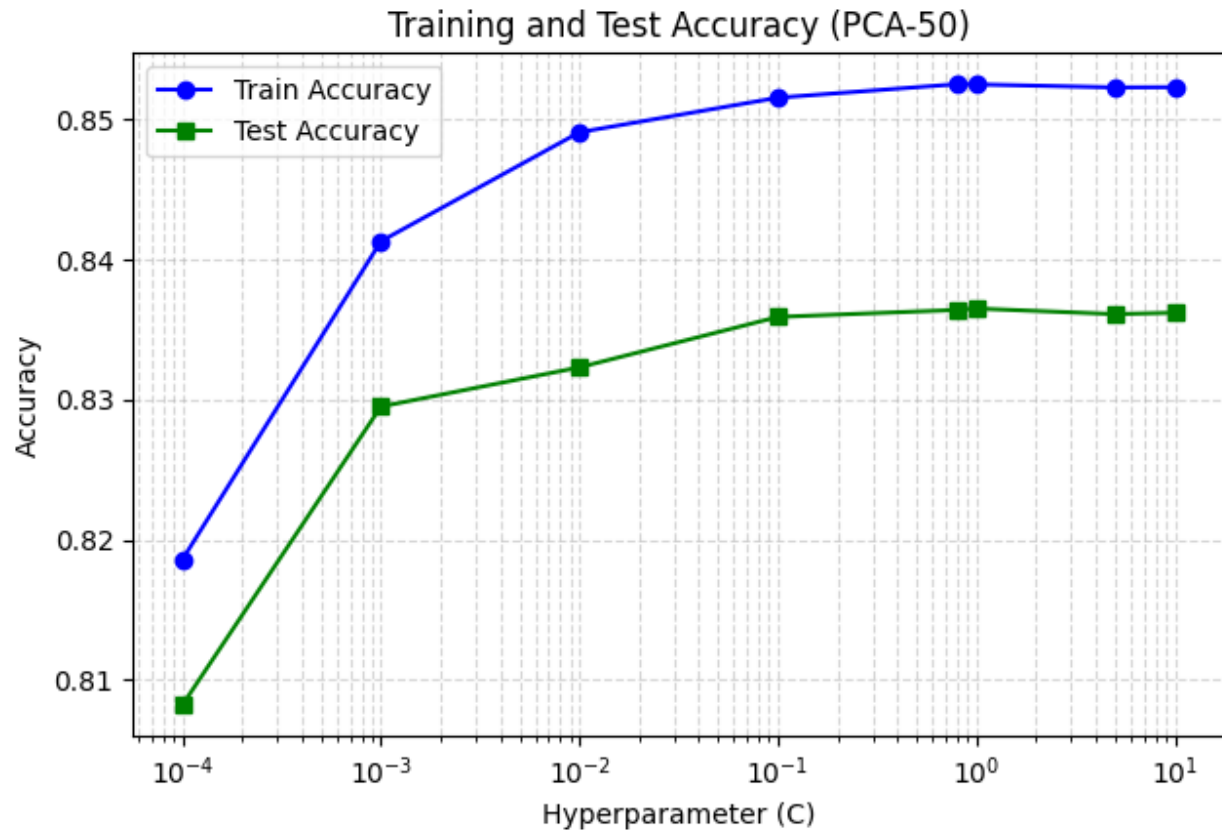
Without dimensionality reduction



C	Training Time	Training Accuracy	Test Accuracy
0.001	513.679502	0.876200	0.8543
0.010	434.367323	0.891683	0.8571
0.100	567.046639	0.904933	0.8456
1.000	939.134082	0.913617	0.8370

C=0.01 looks best with highest test accuracy and lesser training time compared to other values of C.

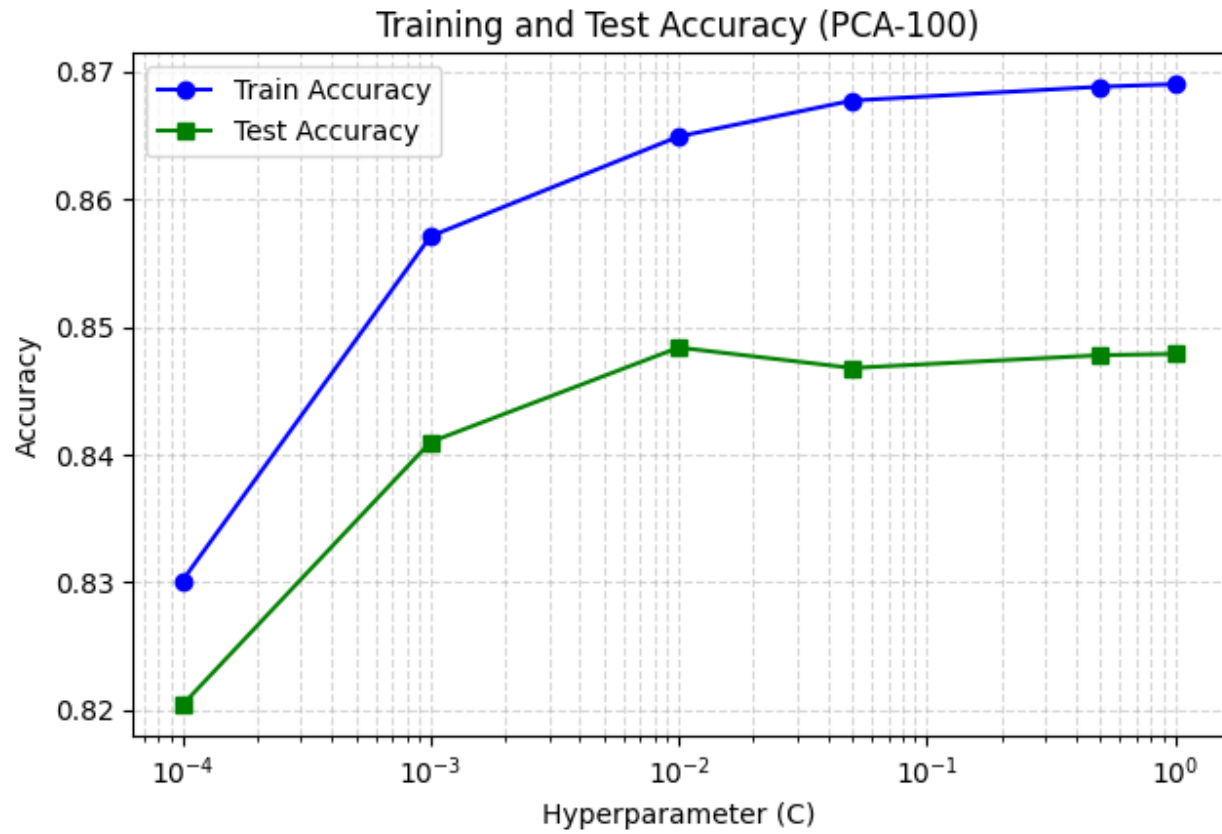
PCA (Dim = 50)



C	Training Time	Training Accuracy	Test Accuracy
0.0001	117.524936	0.818583	0.8082
0.0010	79.523151	0.841300	0.8295
0.0100	73.351838	0.849083	0.8323
0.1000	102.688297	0.851550	0.8359
0.8000	237.034440	0.852517	0.8364
1.0000	269.458771	0.852500	0.8365
5.0000	818.639958	0.852283	0.8361
10.0000	1465.624529	0.852300	0.8362

Taking test accuracy, the gap between test and train accuracy and the training time into consideration, C=0.1 is optimal.

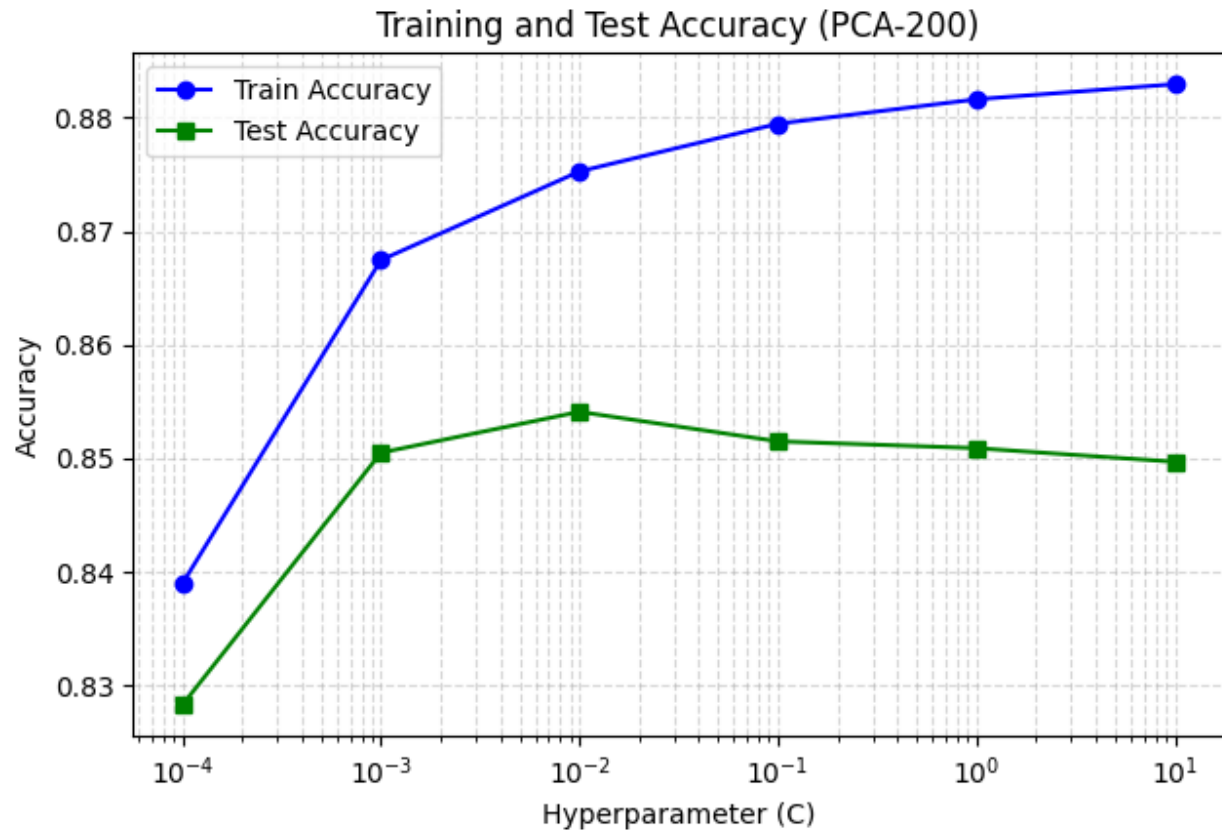
PCA (Dim = 100)



C	Training Time	Training Accuracy	Test Accuracy
0.0001	145.948215	0.830133	0.8204
0.0010	96.167563	0.857117	0.8410
0.0100	89.372633	0.864933	0.8484
0.0500	111.882486	0.867750	0.8468
0.5000	238.809998	0.868833	0.8478
1.0000	361.532346	0.869033	0.8479

C=0.01 gives the optimal outcome.

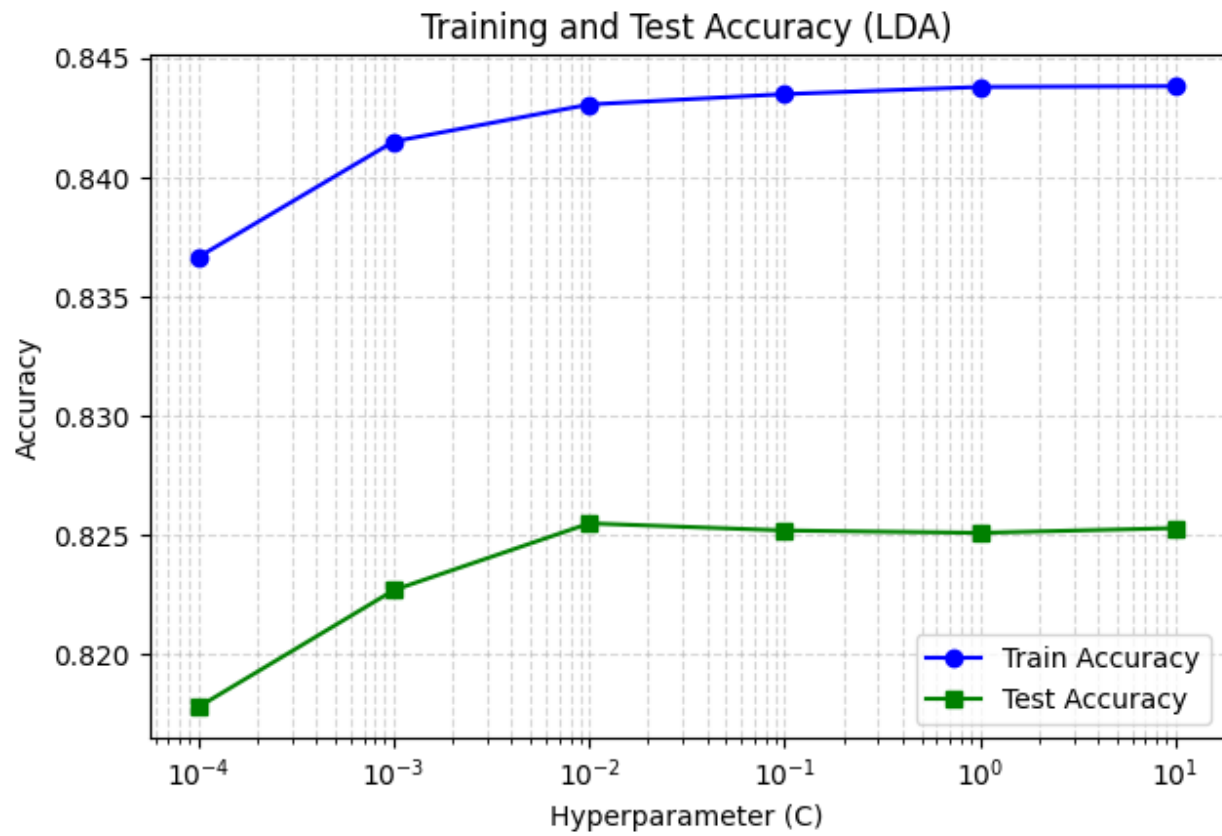
PCA (Dim = 200)



C	Training Time	Training Accuracy	Test Accuracy
0.0001	194.273620	0.838983	0.8283
0.0010	131.902093	0.867450	0.8505
0.0100	118.506508	0.875250	0.8541
0.1000	158.525287	0.879450	0.8515
1.0000	478.747885	0.881600	0.8509
10.0000	3311.261016	0.882933	0.8497

C=0.01 has the highest test accuracy, takes less time for training, and has a reasonable gap with train accuracy.

LDA

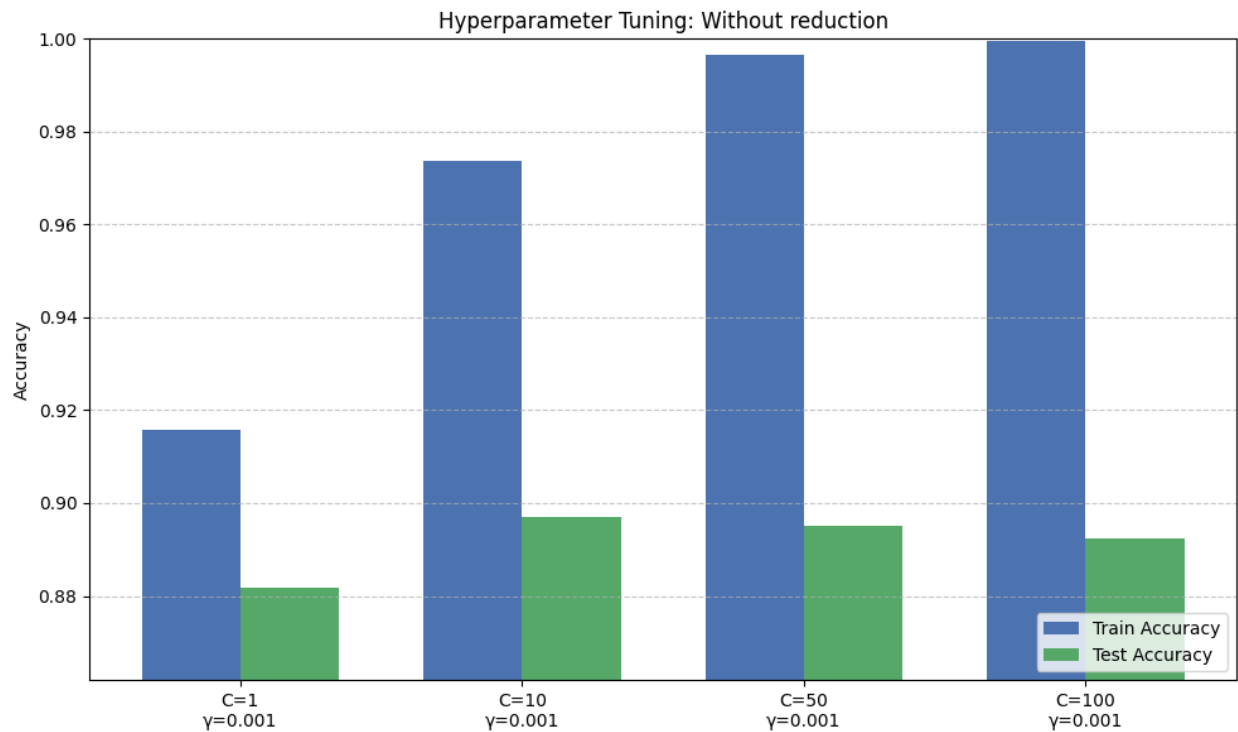


C	Training Time	Training Accuracy	Test Accuracy
0.0001	76.176613	0.836667	0.8178
0.0010	42.457076	0.841517	0.8227
0.0100	34.460866	0.843067	0.8255
0.1000	34.351144	0.843500	0.8252
1.0000	49.125244	0.843800	0.8251
10.0000	103.875039	0.843833	0.8253

C=0.01 is the optimal hyperparameter.

RBF SVC Tuning

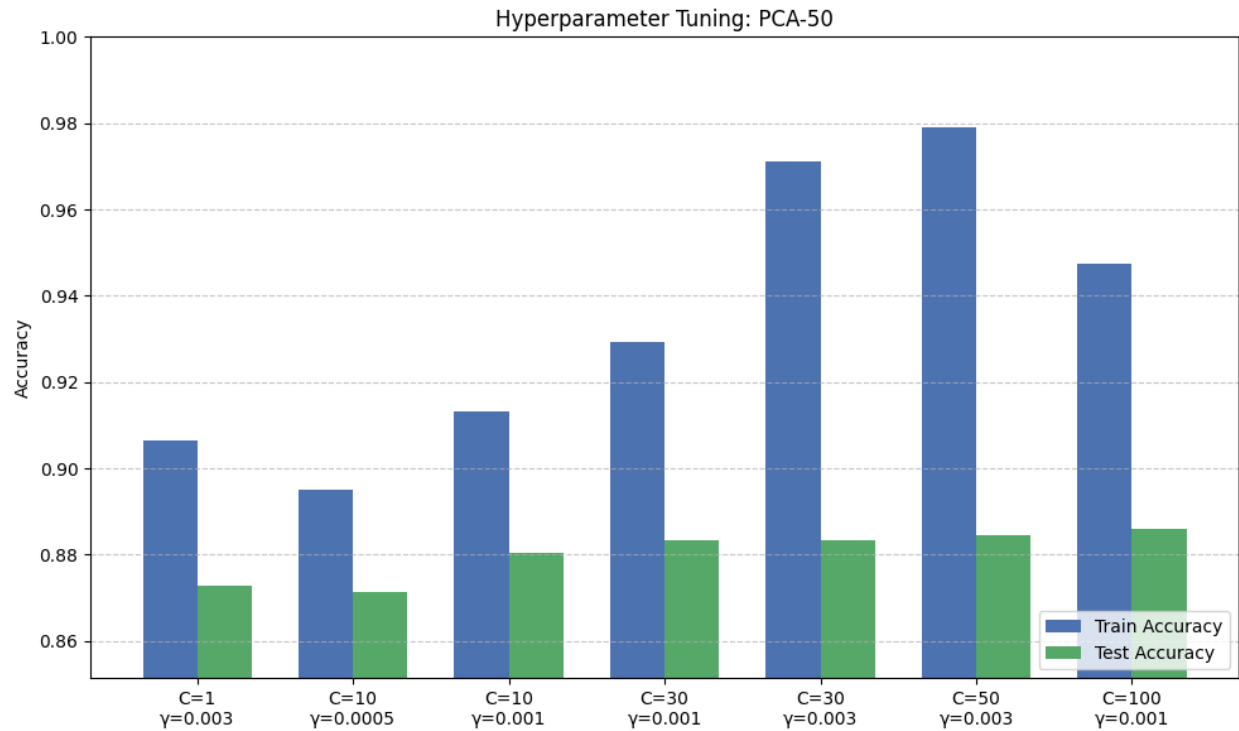
Without dimensionality reduction



C	Gamma	Training Time	Training Accuracy	Test Accuracy
1	0.001	630.131119	0.915767	0.8819
10	0.001	304.279264	0.973583	0.8970
50	0.001	549.024052	0.996433	0.8951
100	0.001	538.702636	0.999483	0.8925

C=10 and Gamma=0.001 is the best hyperparameter.

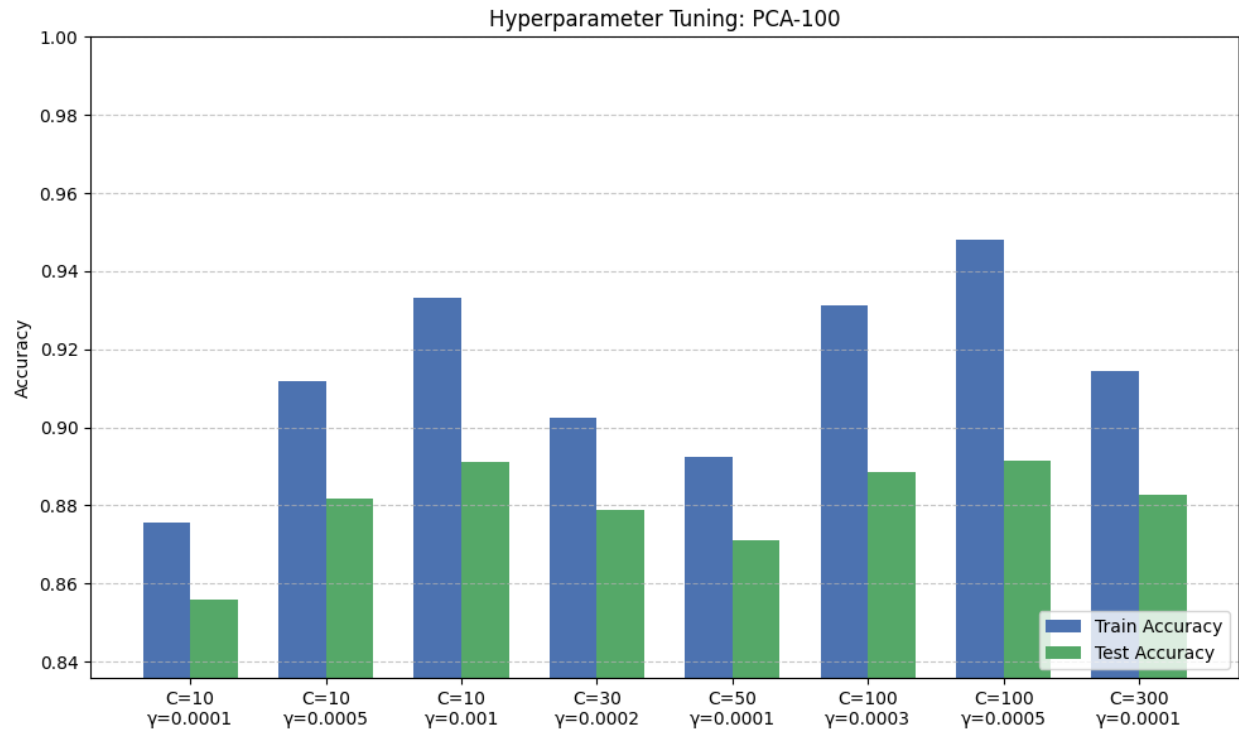
PCA (Dim = 50)



C	Gamma	Training Time	Training Accuracy	Test Accuracy
1	0.0030	102.735107	0.906383	0.8729
10	0.0005	74.702870	0.895017	0.8714
10	0.0010	73.430530	0.913100	0.8804
30	0.0010	73.915243	0.929417	0.8834
30	0.0030	99.224415	0.971117	0.8832
50	0.0030	102.143711	0.979033	0.8844
100	0.0010	86.151545	0.947483	0.8861

C=100 and Gamma=0.001 gives the optimal outcome.

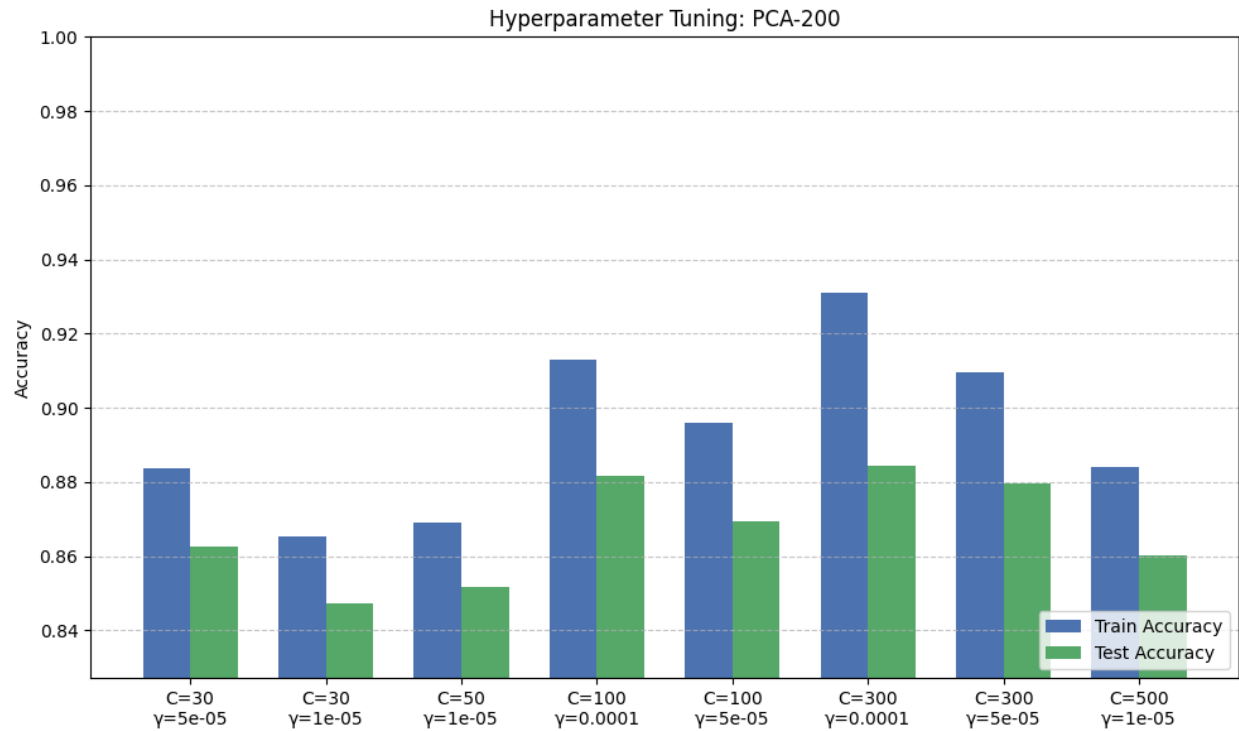
PCA (Dim = 100)



C	Gamma	Training Time	Training Accuracy	Test Accuracy
10	0.0001	96.097785	0.875500	0.8558
10	0.0005	76.085020	0.911833	0.8816
10	0.0010	78.565434	0.933250	0.8912
30	0.0002	76.809795	0.902317	0.8788
50	0.0001	81.731357	0.892567	0.8712
100	0.0003	78.810638	0.931217	0.8886
100	0.0005	81.387249	0.948050	0.8916
300	0.0001	86.791457	0.914300	0.8827

C=10 and Gamma=0.001 is the best hyperparameter.

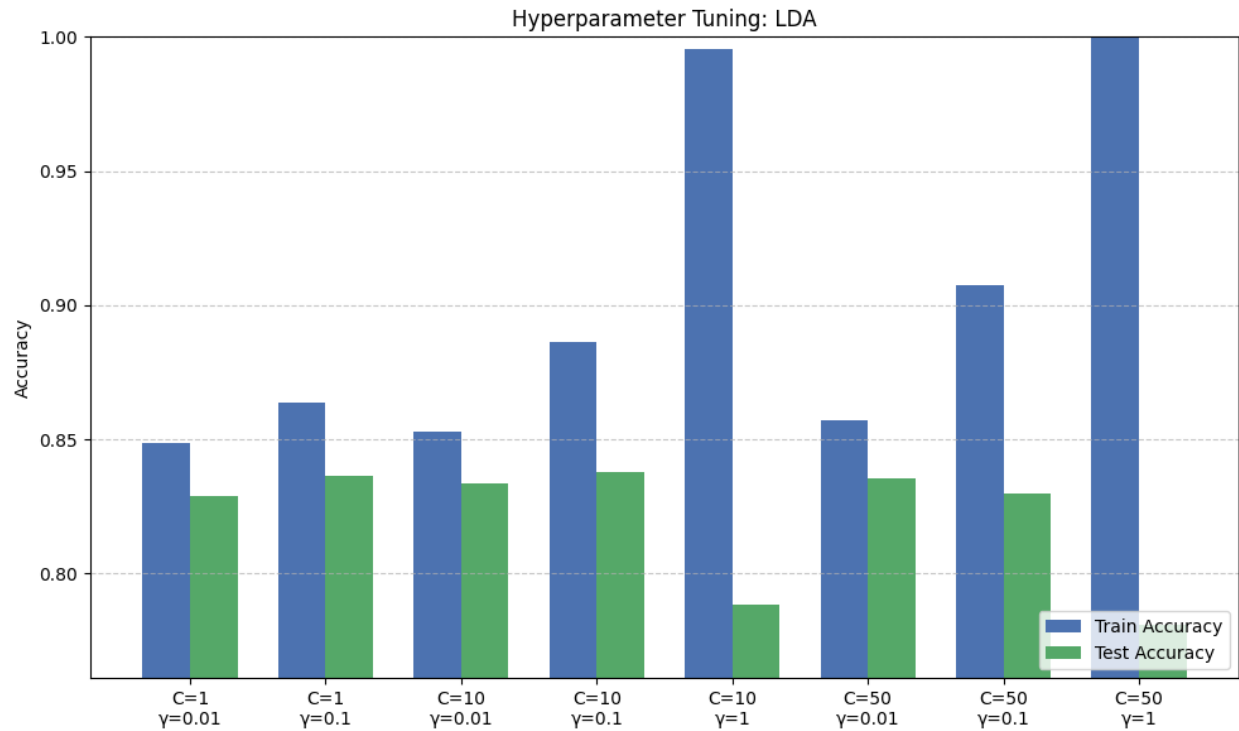
PCA (Dim = 200)



C	Gamma	Training Time	Training Accuracy	Test Accuracy
30	0.00005	117.994272	0.883833	0.8625
30	0.00001	152.320138	0.865283	0.8472
50	0.00001	146.173798	0.869133	0.8518
100	0.00010	115.634121	0.913117	0.8816
100	0.00005	120.364169	0.895983	0.8693
300	0.00010	126.282029	0.930933	0.8843
300	0.00005	123.087858	0.909567	0.8797
500	0.00001	127.878625	0.883983	0.8601

C=100 and Gamma=0.0001 is the best hyperparameter.

LDA

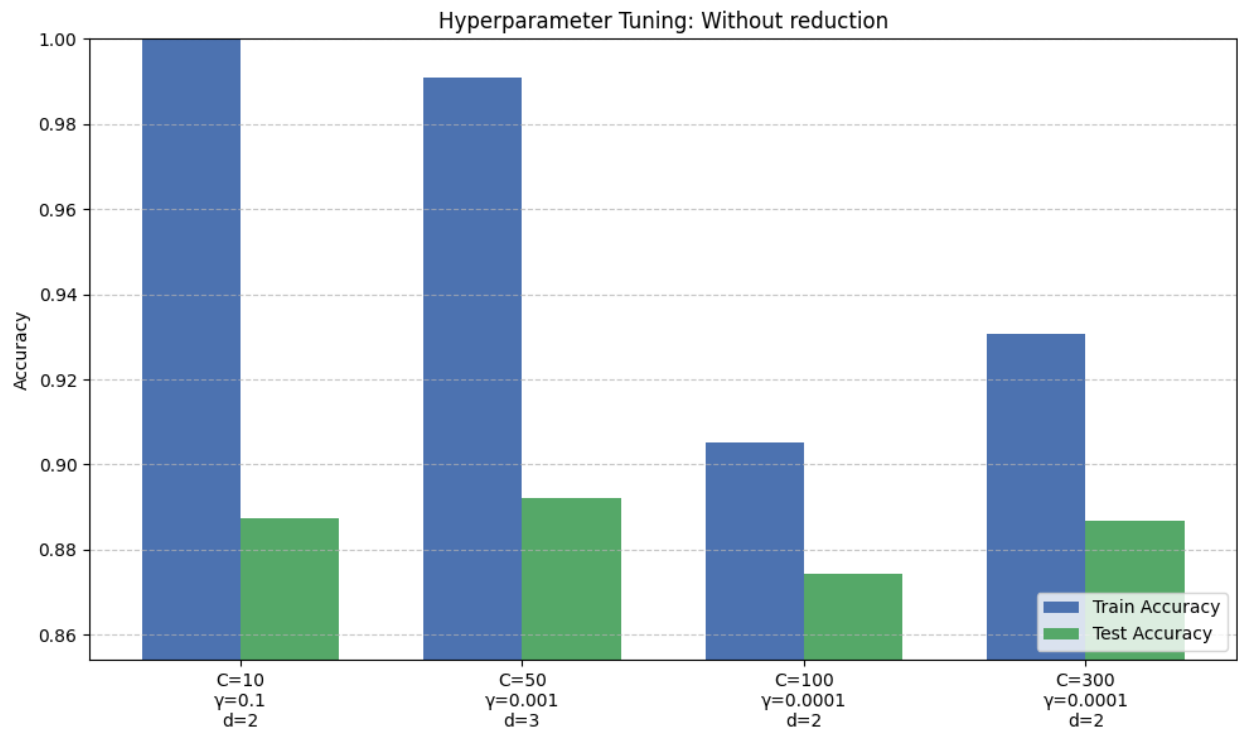


C	Gamma	Training Time	Training Accuracy	Test Accuracy
1	0.01	41.309338	0.848617	0.8289
1	0.10	39.700832	0.863633	0.8362
10	0.01	37.344690	0.852683	0.8335
10	0.10	42.519414	0.886067	0.8377
10	1.00	251.408656	0.995400	0.7884
50	0.01	40.844745	0.856950	0.8354
50	0.10	61.247635	0.907467	0.8298
50	1.00	251.538080	0.999683	0.7811

C=50 and Gamma=0.01 is the best hyperparameter.

Polynomial SVC

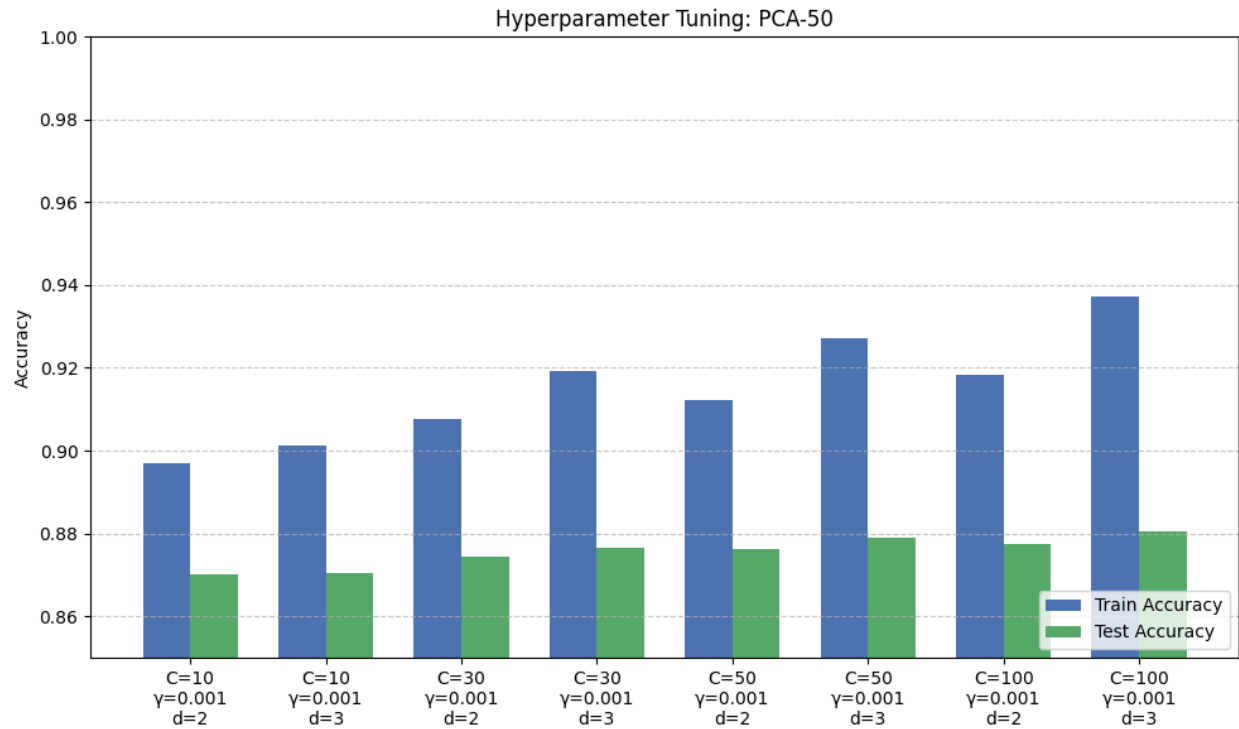
Without dimensionality reduction



C	Gamma	Degree	Training Time	Training Accuracy	Test Accuracy
10	0.1000	2	332.928609	1.000000	0.8875
50	0.0010	3	367.999291	0.990867	0.8921
100	0.0001	2	445.774808	0.905117	0.8742
300	0.0001	2	367.951714	0.930783	0.8869

C=50, Gamma=0.001, and Degree=3 is the best hyperparameter.

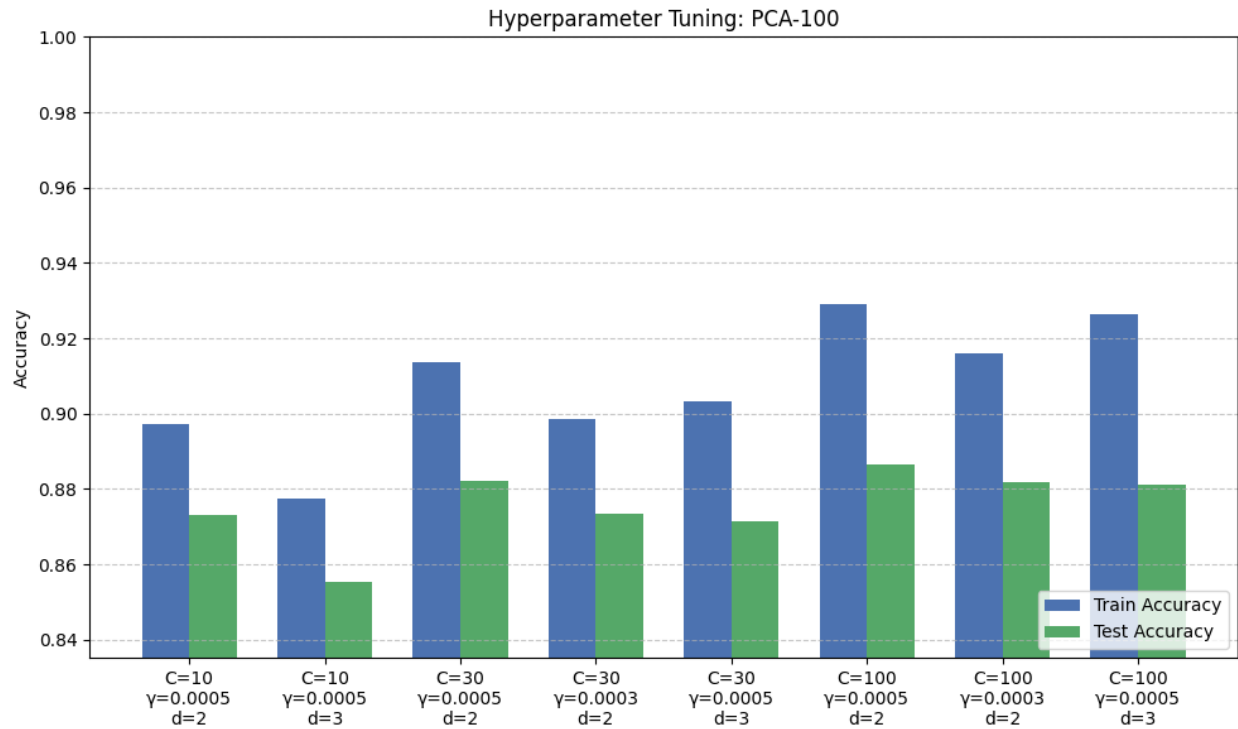
PCA (Dim = 50)



C	Gamma	Degree	Training Time	Training Accuracy	Test Accuracy
10	0.001	2	83.283952	0.896983	0.8700
10	0.001	3	84.870082	0.901267	0.8703
30	0.001	2	75.305117	0.907600	0.8745
30	0.001	3	77.967692	0.919383	0.8764
50	0.001	2	79.034470	0.912217	0.8763
50	0.001	3	74.235631	0.927050	0.8790
100	0.001	2	80.717560	0.918383	0.8773
100	0.001	3	76.400959	0.937150	0.8806

C=50, Gamma=0.001, and Degree=3 is the best hyperparameter.

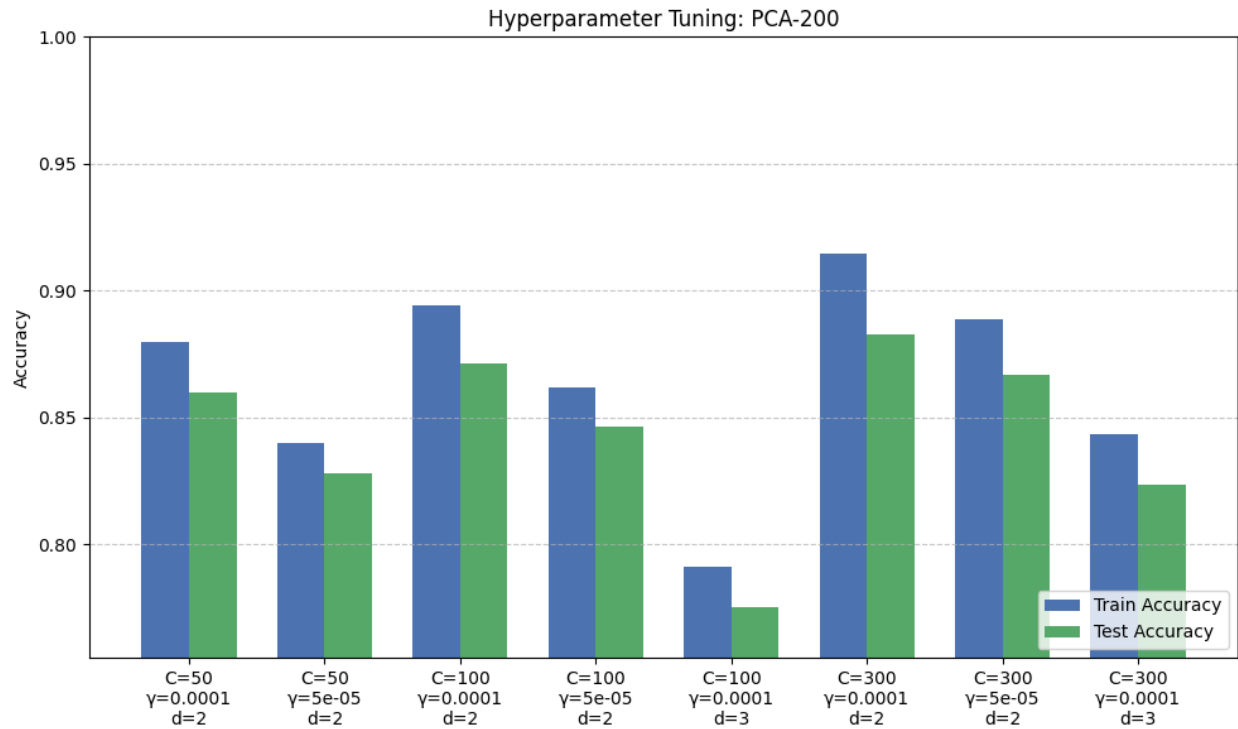
PCA (Dim = 100)



C	Gamma	Degree	Training Time	Training Accuracy	Test Accuracy
10	0.0005	2	125.307288	0.897083	0.8730
10	0.0005	3	167.165664	0.877517	0.8553
30	0.0005	2	94.328116	0.913583	0.8821
30	0.0003	2	106.322854	0.898683	0.8735
30	0.0005	3	111.222120	0.903333	0.8713
100	0.0005	2	83.732119	0.928950	0.8865
100	0.0003	2	87.388631	0.915967	0.8817
100	0.0005	3	94.082414	0.926267	0.8812

C=100, Gamma=0.0005, and Degree=2 is the best hyperparameter.

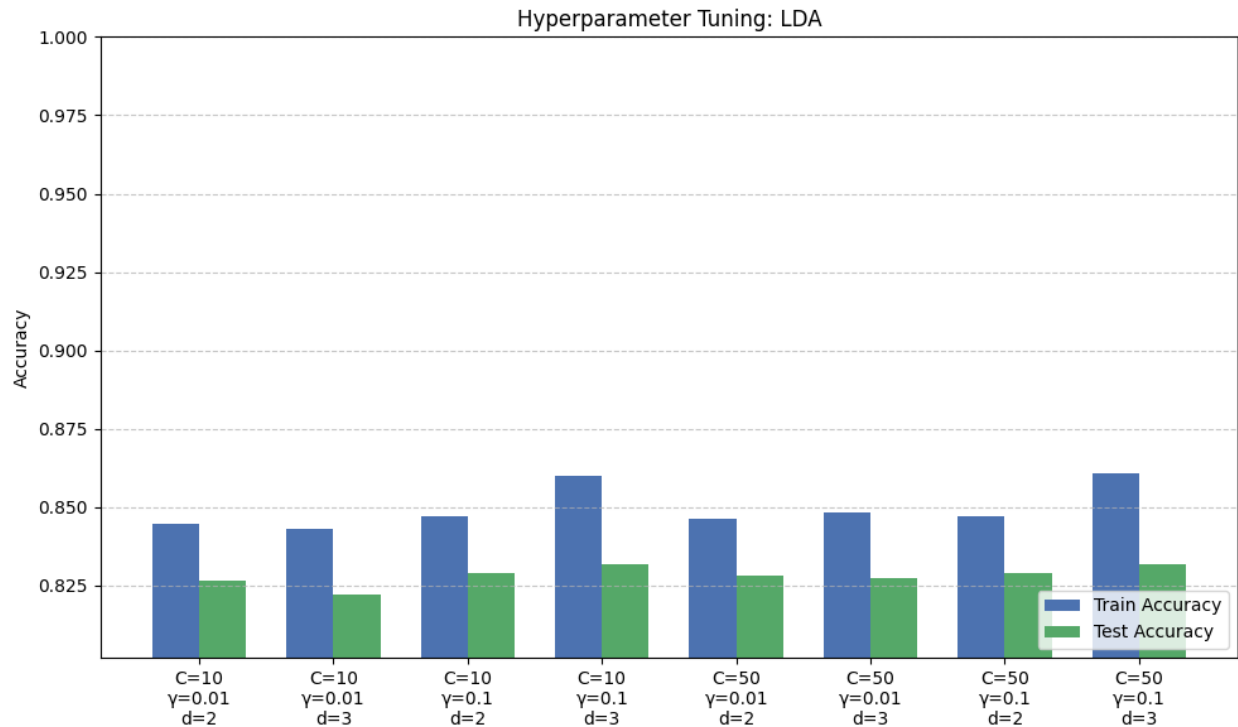
PCA (Dim = 200)



C	Gamma	Degree	Training Time	Training Accuracy	Test Accuracy
50	0.00010	2	224.460629	0.879917	0.8598
50	0.00005	2	322.735573	0.839700	0.8278
100	0.00010	2	222.114725	0.894050	0.8711
100	0.00005	3	235.232669	0.861983	0.8461
100	0.00010	3	406.744575	0.790950	0.7753
300	0.00010	2	125.699843	0.914617	0.8828
300	0.00005	2	166.176219	0.888683	0.8669
300	0.00010	3	272.041563	0.843417	0.8236

C=300, Gamma=0.0001, and Degree=2 is the best hyperparameter.

LDA



C	Gamma	Degree	Training Time	Training Accuracy	Test Accuracy
10	0.01	2	41.813623	0.844583	0.8268
10	0.01	3	39.553451	0.843200	0.8221
10	0.10	2	62.359188	0.847217	0.8290
10	0.10	3	119.768943	0.860050	0.8318
50	0.01	2	34.762874	0.846383	0.8283
50	0.01	3	42.258464	0.848283	0.8274
50	0.10	2	160.157824	0.847167	0.8289
50	0.10	3	474.724183	0.860733	0.8317

C=10, Gamma=0.1, and Degree=2 is the best hyperparameter.

Fashion MNIST ensembling

Training time: 15.303679300002841 s

Accuracy: 0.862, Error: 0.138

MNIST ensembling

Training time: 15.5 s

Accuracy: 0.9854, Error: 0.0146